

**SIR SYED UNIVERSITY OF ENGINEERING & TECHNOLOGY COMPUTER
SCIENCE & INFORMATION TECHNOLOGY DEPARTMENT**

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Economics & Management (HS-412)
Assignment # 1**

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Name: Muzamil Mehmood

Roll #: 2022F-BCS-316

Section: E

Question: Relate linkage between Money - Time relationship. Identify reasons that lead to devaluation with passage of time.

1. Linkage between Money and Time (Money–Time Relationship)

The money–time relationship refers to how the value of money changes over time.

In simple terms : A rupee today is worth more than a rupee tomorrow.

This concept is known as the Time Value of Money (TVM) — it means that money has earning potential over time.

If you have money now, you can invest it and earn interest or profit, but money received in the future cannot be used or invested today.

Key Ideas

- **Present Value (PV):** Current worth of future money.
- **Future Value (FV):** Value of money after earning interest or return over time.
- **Interest Rate (r):** Measures how fast money grows over time.
- **Formula:** $FV = PV \times (1+r)^t$

So, the longer the time or the higher the interest rate, the greater the future value — but also the more inflation may reduce real purchasing power.

2. Reasons for Devaluation of Money Over Time

Money **loses value** over time mainly due to **inflation** and other **economic factors**.

Here are the main reasons:

Reason	Explanation
Inflation	As prices of goods and services increase, each unit of currency buys fewer goods. This reduces purchasing power.
Excess Money Supply	When a government prints too much money, it increases supply without matching economic output — leading to inflation and devaluation.
Fiscal Deficits	Large budget deficits financed through borrowing or money printing reduce currency confidence.
Interest Rate Differences	Lower interest rates reduce foreign investment, weakening the currency.

Trade Deficits	When imports exceed exports, demand for foreign currency rises and domestic currency value falls.
Political or Economic Instability	Uncertainty discourages investment and confidence, lowering currency value.
Depreciation in Foreign Exchange Market	Market forces (supply and demand of currencies) can reduce a currency's value relative to others.

In Summary

- **Link:** Money and time are linked because the *value of money changes over time* — it can grow through investment but also lose value through inflation.
- **Devaluation Causes:** Inflation, money oversupply, fiscal imbalance, low interest rates, trade deficits, and instability reduce the real worth of money with time.